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The Editors
JMIR Mental Health

Dear Editor,

We are pleased to submit our original research manuscript, **"Feature Vectors and Narrative Embeddings for Predicting a Treatment Switching Proxy for Treatment Resistant Depression: Retrospective Cohort Study,"** for consideration in *JMIR Mental Health*.

The study asks whether pretrained embeddings of template-generated patient narratives predict a treatment-switching proxy for treatment-resistant depression (TRD) better than structured feature vectors built from the same electronic health records (EHRs). Prediction is made at the index antidepressant prescription.

In 42,579 patients from one community health system, narrative embeddings did not demonstrate superior discrimination over structured feature vectors. Embedded logistic regression reached a receiver operating characteristic area under the curve (ROC AUC) of 0.657 (95% CI 0.643–0.672). Feature-vector XGBoost reached 0.649 (95% CI 0.634–0.664). Their paired difference was 0.008 (95% CI −0.003 to 0.019). The classifier changed the result: embeddings improved logistic regression and reduced discrimination for all three tree ensembles. Logistic regression led the embedded models on all four encoders, at 0.645 (95% CI 0.629–0.660) to 0.657 (95% CI 0.643–0.672).

The predictive signal came from familiar clinical history. Permuting psychiatric history caused the largest loss, and medication burden and prior treatment also contributed. Permuting race/ethnicity or social determinants changed ROC AUC by no more than 0.003, and every such interval included zero. Nearest-neighbor retrieval over the embedding showed that proximity tracks outcome risk, but it stayed below both leading classifiers at every neighborhood size from 1 to 34,063, for all four encoders. Its maximum for Qwen3-Embedding-8B at similarity exponent 1, 0.625 (95% CI 0.610–0.641), was read at a size chosen on the test patients. Randomly chosen neighbors stayed at chance, at most 0.500 (2.5th–97.5th percentile across 1,000 draws 0.484–0.515).

We are explicit that the outcome is a treatment-switching proxy rather than confirmed treatment resistance, that validation is internal, and that the subgroup analyses do not establish equitable performance. Both representations encode the same hand-selected predictors, so the comparison says nothing about embedding an unselected record. We think the results give a practical structured-feature benchmark for EHR embedding pipelines, and a clear case for validating the outcome

before deploying either approach.

This work is reported in accordance with the **TRIPOD+AI** guideline for the reporting of clinical prediction models that use regression or machine learning methods; a completed TRIPOD+AI checklist is included with this submission.

We confirm that this manuscript describes original work, is not under consideration for publication elsewhere, and that all listed authors have read and approved the submission. The EHR data are not publicly shareable; the full analysis code is available at https://github.com/Pirate-Hunter-Zoro/Atlas/tree/da44a79a7a68f7e57de2e2ba2e27a59835de9530/projects/TRD-EHR. Ethics/IRB, funding, and conflict of interest statements are provided in the manuscript's Ethical Considerations, Funding, and Conflicts of Interest sections: no conflicts of interest are declared; the work was funded by the William K. Warren Foundation, with no other funding source; and the study is a secondary analysis of de-identified data and is therefore not human-subjects research subject to institutional review board approval. All analysis was performed on local institutional hardware with a de-identified extract without direct identifiers, with no patient data transmitted to any external or commercial service.

Thank you for considering our work.

Sincerely,

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on behalf of all authors